



Monetchat Infrastructure Cost Planning Document

1. Project Overview

Monetchat is a visual AI marketplace system consisting of:

- Application server (Next.js + API)
- Background worker
- PostgreSQL database
- Qdrant vector database
- Redis (cache & queue)
- External AI inference (RunPod with Ollama)

2. Architecture Options

◆ **OPTION 1 — Single Server (Recommended for MVP)**

Architecture

One Linux server hosting:

- App
- Worker
- PostgreSQL
- Qdrant
- Redis

External:



- RunPod (AI models)

Infrastructure

Component	Specification	Monthly Cost
Linux Server (DigitalOcean)	4 vCPU / 8 GB RAM / 160 GB SSD	\$48
RunPod GPU (L4)	\$0.39/hour	\$280
RunPod Storage	200 GB	\$14

Total Cost

Server:	\$48
RunPod GPU:	\$280
Storage:	\$14

TOTAL:	~\$342/month

Notes

- Best for MVP and early users
- Simple deployment
- Minimal DevOps complexity

◆ OPTION 2 — Split Architecture (Balanced Production)

Architecture

Server 1:

- App
- Worker

Server 2:

- PostgreSQL



- Qdrant
- Redis

RunPod:

- AI models

Infrastructure

Component	Specification	Monthly Cost
App Server	4 vCPU / 8 GB	\$48
Data Server	4 vCPU / 8 GB	\$48
RunPod GPU	\$0.39/hour	\$280
RunPod Storage	200 GB	\$14

Total Cost

App Server:	\$48
Data Server:	\$48
RunPod GPU:	\$280
Storage:	\$14

TOTAL:	~\$390/month

Notes

- Better performance isolation
- Scales more easily
- Slightly more complex

◆ OPTION 3 — Fully Scalable Production

Architecture

Server 1:

- App

Server 2:

- Worker

Server 3:

- Qdrant

Managed Services:

- PostgreSQL (DigitalOcean Managed DB)
- Redis (Managed)

RunPod:

- AI models

Infrastructure

Component	Specification	Monthly Cost
App Server	4 vCPU / 8 GB	\$48
Worker Server	2 vCPU / 4 GB	\$24
Qdrant Server	4 vCPU / 8 GB	\$48
Managed PostgreSQL	DO Managed DB	\$30–60
Managed Redis	DO Managed Redis	\$15–30
RunPod GPU	\$0.39/hour	\$280
RunPod Storage	200 GB	\$14

Total Cost

Servers:	\$120
Databases:	\$45–90
RunPod GPU:	\$280
Storage:	\$14



TOTAL: ~\$460 - \$520/month

Notes

- Production-grade architecture
 - Highly scalable
 - Higher cost and complexity
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◆ OPTION 4 — Full Self-Hosted (Not Recommended)

Architecture

Everything hosted on your own infrastructure including AI models.

Infrastructure

Component	Specification	Monthly Cost
GPU Server	L4 / A100 equivalent	\$400–\$1000
App + DB Servers	Multiple	\$100+

Total Cost

TOTAL: ~\$500 - \$1200/month

Notes

- High complexity
 - Requires GPU management
 - Not suitable for current stage
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3. RunPod Cost Breakdown



Current Configuration

Resource	Value
GPU	L4
vCPU	16
RAM	94 GB
Storage	200 GB

Pricing

GPU:	\$0.39/hour
Monthly:	~\$280
Storage:	\$14
TOTAL:	~\$294/month

4. Cost Optimization Opportunities

Reduce RunPod Cost

Method	Impact
Auto-stop idle pods	Save 50–70%
Use smaller GPU	Save 20–40%
On-demand startup	Pay only when used

Recommended Optimization Strategy

- Keep GPU OFF during idle periods
 - Start only when users are active
 - Monitor usage before scaling
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5. Final Recommendation

Recommended Architecture



Option 1 — Single Linux Server + RunPod

Final Setup

Linux Server:

- App
- Worker
- PostgreSQL
- Qdrant
- Redis

RunPod:

- Qwen (chat)
 - LLaVA (vision)
 - Nomic (embeddings)
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Estimated Monthly Cost

TOTAL: ~\$340/month

Why This is Recommended

- Lowest complexity
 - Fastest to deploy
 - Sufficient for MVP
 - Easy to scale later
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6. Future Scaling Plan

When user base grows:

1. Move Qdrant to separate server
 2. Move PostgreSQL to managed DB
 3. Add load balancer
 4. Scale app horizontally
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7. Summary

Option	Cost	Complexity	Recommendation
Single Server	~\$340	Low	✅ Best now
Split Servers	~\$390	Medium	Next stage
Full Production	~\$500	High	Later
Self-host AI	\$500+	Very High	❌ Not needed

If you want, I can convert this into a **proper PDF layout or investor-ready version** next.